

Diagonal Gradient Extension and Otsu-Based Evaluation for Lackadaisical Quantum Walk Edge Detection

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by

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June, 2026

Declaration

I, **Anish Chattopadhyay**, declare that this thesis “*Diagonal Gradient Extension and Otsu-Based Evaluation for Lackadaisical Quantum Walk Edge Detection*” is a bona fide record of the work carried out by me under the supervision of **Mr. Shambo Chatterjee, Assistant Professor**. This report represents my ideas in my own words and where others’ ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea, data, fact, or source in my submission. I further declare that the work reported in this thesis has not been and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or any other institute. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Certificate

This is to certify that the thesis entitled

“Diagonal Gradient Extension and Otsu-Based Evaluation for Lackadaisical Quantum Walk Edge Detection”

submitted by **Anish Chattopadhyay** (Reg. No.: 020721, Roll: 90/MCA No.: 240003) in partial fulfillment for the award of the degree of **Master of Computer Applications in Computer Science and Engineering** from the **University of Kalyani**, is a bona fide work carried out under my supervision during the academic year 2025–2026.

The thesis fulfills the requirements as per the regulations of this University and meets the necessary standards for submission. The contents have not been submitted for the award of any other degree or diploma of this or any other institute.

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Abstract

Edge detection is a fundamental task in digital image processing, used to identify pixel-level boundaries between regions of differing intensity. Classical methods such as Sobel and Canny examine each pixel individually, making them slow for large images and sensitive to noise.

Quantum edge detection methods like QSobel and Hadamard Edge Detection (HED) offer speedup but suffer from critically low measurement success probabilities, limiting their reliability. The lackadaisical discrete-time quantum walk search (DTQWS) method of Giri et al. (2025) addresses this by treating edge detection as a quantum spatial search problem, achieving success probabilities as high as $p_s \approx 0.98$.

This thesis proposes two contributions extending Giri et al. (2025). First, the gradient oracle (Eq. 3) is extended from four axis-aligned directions to six by adding two downward diagonal neighbours ($\downarrow\rightarrow$ and $\downarrow\leftarrow$), allowing the marked set $\mathcal{T}_M$ to capture diagonal edges missed by the original oracle. Second, Otsu’s histogram-based thresholding replaces the paper’s unspecified post-processing step to convert the quantum walk probability map into a binary edge image without relying on any ground-truth information.

Experiments on 20 BSDS500 images show consistent improvement over the baseline across all three evaluation metrics: Raw Oracle F1 ($0.2571 \rightarrow 0.2598$), Otsu Oracle F1 ($0.2554 \rightarrow 0.2664$), and Otsu Quantum F1 ($0.2572 \rightarrow 0.2602$). An ablation study across downward, upward, and full 8-direction diagonal variants confirms that the downward diagonal extension performs best, attributed to better alignment with BSDS500 ground-truth boundary annotations. A Qiskit-based block circuit implementation is also developed and verified against exact statevector simulation.

Keywords: Quantum walk search, Edge detection, Phase oracle, Quantum Edge Detection, Lackadaisical quantum walk, NEQR, BSDS500, Quantum image processing.

List of Abbreviations

Abbreviation	Full Form
BSDS500	Berkeley Segmentation Dataset 500
DTQWS	Discrete-Time Quantum Walk Search
F1	F1 Score — Harmonic Mean of Precision and Recall
FRQI	Flexible Representation of Quantum Images
GT	Ground Truth
HED	Hadamard Edge Detection
MCA	Master of Computer Applications
MPS	Matrix Product State
NEQR	Novel Enhanced Quantum Representation
NISQ	Noisy Intermediate-Scale Quantum
ODS	Optimal Dataset Scale
PEC	Probabilistic Error Cancellation
QED	Quantum Edge Detection
QPIE	Quantum Probability Image Encoding
QW	Quantum Walk
QWS	Quantum Walk Search
ZNE	Zero-Noise Extrapolation
2D	Two Dimensional

List of Symbols

Symbol	Description
<i>Image and Gradient Symbols</i>	
$I(x, y)$	Intensity of pixel at position (x, y)
N	Total number of pixels, $N = N_1 \times N_2$
N_1, N_2	Height and width of the image in pixels
M	Number of marked edge pixels
$G_{\max}(x, y)$	Extended 6-direction gradient at pixel (x, y)
$I_{\max}(x, y)$	Original 4-direction gradient at pixel (x, y)
a_{th}	Gradient threshold for edge marking
$P(x, y)$	Quantum walk probability map at pixel (x, y)
$E(x, y)$	Binary edge map after Otsu thresholding
<i>Quantum State and Operator Symbols</i>	
$ \psi_{in}\rangle$	Initial quantum state
$ \psi_{best}\rangle$	Quantum state at peak success probability
$\mathcal{H}_V$	Vertex Hilbert space
$\mathcal{H}_C$	Coin Hilbert space
U	Evolution operator, $U = S \cdot \mathcal{C}_G$
S	Flip-flop shift operator
$\mathcal{C}_G$	Modified Grover coin operator
$\mathcal{T}_M$	Set of marked edge pixels
<i>Parameter Symbols</i>	
s	Self-loop weight parameter
t	Number of quantum walk iterations
t^*	Peak iteration (iteration of maximum p_s)
T	Maximum number of iterations
p_s	Success probability of quantum walk
$\bar{p}_s$	Average success probability over all blocks
t_{Otsu}^*	Otsu threshold computed from signal histogram
θ_i	FRQI encoding angle for pixel i

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Chapter 1

Introduction

1.1 Background and Motivation

Edge detection identifies pixel-level boundaries between regions of differing intensity in an image and serves as a critical preprocessing step in medical imaging, autonomous vehicles, forensic analysis, and remote sensing. Classical methods such as Sobel [15], Prewitt [14], and Canny [2] compute intensity gradients and apply a threshold to classify pixels as edge or non-edge. While effective on small images, these methods process each pixel sequentially, making them computationally expensive as image sizes grow.

Quantum computing exploits superposition, entanglement, and quantum parallelism to achieve speedups over classical algorithms. Quantum edge detection methods such as QSobel [20] and Hadamard Edge Detection (HED) [19] encode pixel intensities as quantum amplitudes, offering exponential speedup in circuit depth. However, both methods suffer from critically low measurement success probabilities, making reliable edge detection difficult in noisy quantum environments. The lackadaisical discrete-time quantum walk search (DTQWS) of Giri et al. [6] addresses this by treating edge detection as a quantum spatial search problem, achieving success probabilities as high as $p_s \approx 0.98$ compared to QSobel ($p_q \approx 0.0095$) and HED ($\bar{p}_h \approx 0.0031$).

1.2 Problem Statement

Despite its high success probability, the DTQWS method of Giri et al. [6] has two limitations. First, its gradient oracle (Eq. 3) computes gradients only in four axis-aligned directions, meaning diagonal edges may fall below the threshold and remain undetected. Second, the paper does not specify how the quantum walk’s continuous probability map should be converted into a binary edge image, making the evaluation protocol undefined and difficult to reproduce.

1.3 Objectives of the Project

1. Faithfully reproduce the DTQWS algorithm of Giri et al. on 20 BSDS500 images and verify that success probabilities match the paper’s reported results.
2. Extend the gradient oracle from four axis-aligned directions to six by adding two downward diagonal neighbours, enabling detection of diagonal edges missed by the original oracle.

3. Propose Otsu’s histogram-based thresholding as a principled, ground-truth-free method for converting the probability map into a binary edge image.
4. Conduct an ablation study comparing downward, upward, and 8-direction diagonal variants to identify the best configuration.
5. Implement and verify a Qiskit block circuit for the DTQWS on AerSimulator and evaluate it across $t = 1$ to $t = 5$ iterations.

1.4 Scope of the Project

All experiments use classical statevector simulation of the quantum walk in numpy — the full two-dimensional walk is computed via exact matrix operations, without shot noise or hardware errors. Evaluation is conducted exclusively on 20 images from the BSDS500 dataset [10] using Precision, Recall, and F1 score under fixed-threshold (Raw Oracle) and Otsu adaptive thresholding protocols. A Qiskit circuit implementation follows the paper’s 2×2 block decomposition and is validated on the AerSimulator. Real quantum hardware execution and deep learning comparisons are outside the scope of this work.

1.5 Challenges in Current Quantum Walk Edge Detection

Current quantum walk edge detection faces four key challenges. First, NISQ hardware limitations prevent full-scale deployment — the complete two-dimensional DTQWS requires circuit depths far beyond current device capabilities. Second, the binary oracle applies an identical phase flip to all marked vertices regardless of gradient strength, causing the walk to amplify weak and strong edges equally. Third, the axis-aligned gradient oracle misses diagonal edges that are common in natural images. Finally, undefined post-processing in the original paper makes reproducible evaluation impossible.

1.6 Proposed Solution Overview

This thesis addresses the last two challenges through two contributions. The **first** extends the gradient oracle to six directions by adding downward diagonal neighbours:

$$G_{\max}(x, y) = \max(|I_{h+}|, |I_{h-}|, |I_{v+}|, |I_{v-}|, |I_{d\downarrow\rightarrow}|, |I_{d\downarrow\leftarrow}|) \quad (1.1)$$

where $I_{d\downarrow\rightarrow}(x, y) = I(x, y) - I(x+1, y+1)$ and $I_{d\downarrow\leftarrow}(x, y) = I(x, y) - I(x+1, y-1)$. The coin and shift operators remain unchanged, preserving the quantum walk’s theoretical guarantees. The **second** adopts Otsu’s method [12] to compute a data-driven threshold t^* from the probability map’s histogram, replacing the paper’s undefined post-processing with a reproducible, ground-truth-free protocol.

Chapter 2

Literature Review

2.1 Classical Edge Detection Methods

Edge detection has been a central problem in computer vision since the early 1970s. The Sobel operator [15] computes horizontal and vertical gradient approximations using 3×3 convolution kernels, and an edge is declared where the gradient magnitude exceeds a threshold. The Prewitt operator [14] uses uniform kernel weights rather than distance-weighted ones, making it slightly less sensitive to diagonal edges. The Kirsch compass masks [7] extend gradient computation to eight orientations by applying eight rotated kernels and taking the maximum response, which improves directional sensitivity but increases computational cost. The Roberts cross operator uses diagonal 2×2 kernels for faster computation but is highly sensitive to noise due to the absence of any smoothing.

The Canny edge detector [2] is widely regarded as the most effective classical method. It operates in four stages: Gaussian smoothing to reduce noise, gradient magnitude and direction computation, non-maximum suppression to thin edges to single-pixel width, and double thresholding with hysteresis to distinguish strong edges from weak ones. The hysteresis step retains weak edges only if they connect to strong ones, significantly reducing false positives. Despite its effectiveness, Canny requires careful parameter tuning and its performance degrades under varying lighting conditions and textured regions. Crucially, all classical methods share a fundamental bottleneck: their time complexity scales as $O(N)$ with the number of pixels N , making them increasingly inadequate for high-resolution, real-time applications [16].

2.2 Quantum Image Representation

For quantum algorithms to process images, pixel intensities must first be encoded into quantum states. The Flexible Representation of Quantum Images (FRQI) [8] encodes a grayscale image of $2^n \times 2^n$ pixels into a normalized quantum state:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (2.1)$$

where $\theta_i \in [0, \pi/2]$ encodes the intensity of the i -th pixel and $|i\rangle$ is the positional basis state. FRQI requires only $2n + 1$ qubits to represent 2^{2n} pixels, offering exponential compression. However, retrieving individual pixel information requires $O(2^{2n})$ operations.

The Novel Enhanced Quantum Representation (NEQR) [21] addresses this limitation by using a qubit sequence to encode intensity values directly as a binary string, enabling constant-

time retrieval of any single pixel and supporting a wider range of image operations. Both FRQI and NEQR allow quantum gradient operations to be applied to all pixels simultaneously through superposition, forming the foundation for quantum edge detection algorithms.

2.3 Quantum Edge Detection Methods

2.3.1 QSobel

Zhang et al. [20] proposed QSobel, a quantum analogue of the classical Sobel operator. The method applies quantum circuits to compute horizontal and vertical gradient operators simultaneously across all pixels encoded in the FRQI representation. The total probability of measuring any edge pixel is bounded by:

$$p_q = \frac{M}{N} \quad (2.2)$$

where M is the number of edge pixels and N is the total pixel count. The actual success probability is even lower: $p_q = \sum_{(x,y) \in \mathcal{T}_M} \sin^2 \theta_{xy} / N$. Since edges are a small fraction of any natural image ($M \ll N$), the success probability is extremely low in practice, making the raw output dominated by background noise and requiring significant classical post-processing to produce a usable edge map.

2.3.2 Hadamard Edge Detection

Yao et al. [19] introduced Hadamard Edge Detection (HED), which encodes pixel intensities as probability amplitudes using Quantum Probability Image Encoding [3]. A Hadamard gate is applied to the first qubit of the encoded state, producing a superposition where odd-indexed basis states encode pairwise pixel differences and even-indexed states are discarded. This design wastes more than half the available probability mass, mathematically limiting the total success probability to:

$$p_h \leq \frac{1}{2} \quad (2.3)$$

In practice, Giri et al. [6] report $\bar{p}_h \approx 0.0031$ on benchmark images — far below this theoretical ceiling — because most adjacent pixel pairs have small differences that contribute negligible probability to the odd-indexed states.

2.3.3 Quantum Walk-Based Methods

Quantum walk [13] is the quantum analogue of classical random walk and has emerged as a powerful framework for spatial search on graphs. Childs and Goldstone [4] showed that continuous-time quantum walk can find a marked vertex on an n -dimensional lattice efficiently, and subsequent discrete-time work on two-dimensional grids [1, 17] established $O(\sqrt{N \log N})$ complexity. The lackadaisical quantum walk [18] attaches self-loops with weight s to each vertex, and Giri [5] showed that adding long-range edges achieves the optimal $O(\sqrt{N})$ complexity even on the two-dimensional grid. These results established quantum walk as a viable mechanism for efficient spatial search, motivating its application to image edge detection.

2.4 Lackadaisical DTQWS Edge Detection

Giri et al. [6] proposed the first quantum edge detection method based on lackadaisical discrete-time quantum walk search. The algorithm places image pixels at vertices of an $N_1 \times N_2$ lattice with four regular edges and one self-loop of weight s per vertex. Edge pixels are identified as marked vertices using the gradient threshold $I_{\max}(x, y) \geq a_{th}$. The quantum walk begins from the uniform superposition initial state:

$$|\psi_{in}\rangle = \frac{1}{\sqrt{N}} \sum_{x,y} |x, y\rangle \otimes \frac{1}{\sqrt{4+s}} (|r\rangle + |l\rangle + |u\rangle + |d\rangle + \sqrt{s}|sl\rangle) \quad (2.4)$$

and evolves under $U = S \cdot \mathcal{C}_G$, where the modified Grover coin $\mathcal{C}_G$ flips the phase of the self-loop component at marked vertices and S is the flip-flop shift operator. This construction avoids exceptional configurations [9, 11] that trap standard coins, allowing the walk to amplify probability at any configuration of marked vertices. The method achieves $p_s \approx 0.98$, dramatically outperforming both QSobel ($p_q \approx 0.0095$) and HED ($\bar{p}_h \approx 0.0031$). For NISQ deployment, the paper implements a four-qubit Qiskit circuit that operates on 2×2 image blocks with $t = 2$ iterations as a proof of concept.

2.5 Limitations of Existing Work and Research Gap

Despite its strong performance, the DTQWS method of Giri et al. [6] leaves two important aspects unaddressed. First, the gradient oracle in Eq. (3) computes gradients only in four axis-aligned directions (horizontal and vertical), ignoring diagonal intensity transitions entirely. Diagonal edges are prevalent in natural images — object boundaries, slanted surfaces, and curved contours all produce strong diagonal gradients that a four-direction oracle may classify as non-edges, directly reducing the recall of the marked set $\mathcal{T}_M$. Second, the paper does not define a reproducible post-processing threshold for converting the quantum walk’s continuous probability map into a binary edge image. A threshold a_{th} is mentioned but never specified, making it impossible to evaluate the method in a consistent, ground-truth-free manner across different images. These two gaps — an incomplete gradient oracle and an undefined output thresholding protocol — form the precise research gaps this thesis addresses.

Chapter 3

Proposed Methodology

This chapter presents the two contributions of this thesis in detail. Section 3.1 provides an overview of the baseline DTQWS algorithm and its mathematical formulation. Section 3.2 describes the six-direction diagonal gradient extension. Section 3.3 presents Otsu-based probability map thresholding. Section 3.4 describes the Qiskit circuit implementation. Sections 3.5, 3.6, and 3.7 discuss the advantages, limitations, and a summary of the proposed approach.

3.1 Baseline: Lackadaisical DTQWS

The proposed method builds directly on the lackadaisical DTQWS of Giri et al. [6]. We briefly restate the mathematical formulation to establish notation and to clearly identify where our contributions modify the baseline.

3.1.1 State Space and Initial State

Let an image have pixel intensities $I(x, y)$ at vertices of an $N_1 \times N_2$ rectangular lattice with $N = N_1 N_2$ vertices. Each vertex has four regular directed edges (right r , left l , up u , down d) and one self-loop sl with weight s . The total Hilbert space is $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_C$ where $\mathcal{H}_V$ is the N -dimensional vertex space and $\mathcal{H}_C$ is the 5-dimensional coin space spanned by $\{|r\rangle, |l\rangle, |u\rangle, |d\rangle, |sl\rangle\}$.

The quantum walk begins from the initial state $|\psi_{in}\rangle = |\psi_v\rangle \otimes |\psi_c\rangle$ where:

$$|\psi_v\rangle = \frac{1}{\sqrt{N}} \sum_{x=1}^{N_1} \sum_{y=1}^{N_2} |x, y\rangle \quad (1)$$

is the uniform superposition over all vertices, and:

$$|\psi_c\rangle = \frac{1}{\sqrt{4+s}} (|r\rangle + |l\rangle + |u\rangle + |d\rangle + \sqrt{s}|sl\rangle) \quad (2)$$

is the initial coin state weighted by the self-loop parameter s .

3.1.2 Gradient Oracle and Marked Set

The original gradient oracle (Eq. 3 of Giri et al.) computes, for each pixel (x, y) , the maximum absolute intensity difference over four axis-aligned neighbours:

$$\begin{aligned} I_{h\pm}(x, y) &= I(x, y) - I(x \pm 1, y) \\ I_{v\pm}(x, y) &= I(x, y) - I(x, y \pm 1) \\ I_{\max}(x, y) &= \max(I_{h+}, I_{h-}, I_{v+}, I_{v-}) \end{aligned} \quad (3)$$

A pixel is marked as an edge and added to the set $\mathcal{T}_M$ if:

$$I_{\max}(x, y) \geq a_{th} \quad (3')$$

where a_{th} is a pre-determined threshold value.

3.1.3 Modified Coin Operator

The modified Grover coin operator $\mathcal{C}_G$ acts on the joint state as:

$$\mathcal{C}_G|x, y\rangle \otimes |b_c\rangle = \begin{cases} C|x, y\rangle \otimes |b_c\rangle & \text{if } (x, y) \notin \mathcal{T}_M \\ C|x, y\rangle \otimes |b_c\rangle & \text{if } (x, y) \in \mathcal{T}_M, b_c \neq sl \\ -C|x, y\rangle \otimes |b_c\rangle & \text{if } (x, y) \in \mathcal{T}_M, b_c = sl \end{cases} \quad (4)$$

where $C = 2|\psi_c\rangle\langle\psi_c| - \mathbb{I}_{5 \times 5}$ is the Grover diffusion operator acting on the coin space. The key feature of this operator is that it flips the phase only of the self-loop component at marked vertices. For unmarked vertices and for all non-self-loop coin components at marked vertices, the standard Grover diffusion is applied unchanged.

3.1.4 Shift Operator and Evolution

The flip-flop shift operator S moves the walker in the direction specified by the coin state and simultaneously flips the coin direction:

$$\begin{aligned} S|x, y\rangle \otimes |r\rangle &= |x+1, y\rangle \otimes |l\rangle \\ S|x, y\rangle \otimes |l\rangle &= |x-1, y\rangle \otimes |r\rangle \\ S|x, y\rangle \otimes |u\rangle &= |x, y+1\rangle \otimes |d\rangle \\ S|x, y\rangle \otimes |d\rangle &= |x, y-1\rangle \otimes |u\rangle \\ S|x, y\rangle \otimes |sl\rangle &= |x, y\rangle \otimes |sl\rangle \end{aligned} \quad (5)$$

The evolution operator is $U = S \cdot \mathcal{C}_G$, applied iteratively for t steps. The success probability after t steps is:

$$p_s = \sum_{(x,y) \in \mathcal{T}_M} \sum_{b_c} |\langle x, y| \otimes \langle b_c| U^t |\psi_{in}\rangle|^2 \quad (6)$$

The walk is run for up to $T = 1000$ steps and the state $|\psi_{best}\rangle$ at the iteration t^* achieving maximum p_s is retained. The final probability map is computed as:

$$P(x, y) = \sum_{b_c} |\langle x, y| \otimes \langle b_c| \psi_{best}\rangle|^2 \quad (3.1)$$

3.2 Six-Direction Diagonal Gradient Extension

3.2.1 Motivation

The four-direction oracle in Eq. (3) computes gradients only along the horizontal and vertical axes. For a pixel lying on a diagonal edge — a boundary that runs at approximately 45 or 135 to the horizontal — the horizontal and vertical differences may both be small even when the true edge contrast is high. This causes such pixels to fall below the threshold a_{th} and remain unmarked, reducing the recall of $\mathcal{T}_M$.

3.2.2 Extended Gradient Oracle

We propose extending the gradient computation to six directions by additionally computing the gradient toward the lower-right ($\downarrow\rightarrow$) and lower-left ($\downarrow\leftarrow$) diagonal neighbours. The extended gradient oracle is defined as:

$$\begin{aligned} I_{d\downarrow\rightarrow}(x, y) &= I(x, y) - I(x + 1, y + 1) \\ I_{d\downarrow\leftarrow}(x, y) &= I(x, y) - I(x + 1, y - 1) \\ G_{\max}(x, y) &= \max(|I_{h+}|, |I_{h-}|, |I_{v+}|, |I_{v-}|, |I_{d\downarrow\rightarrow}|, |I_{d\downarrow\leftarrow}|) \end{aligned} \quad (3^*)$$

A pixel is marked if $G_{\max}(x, y) \geq a_{th}$. Since $G_{\max}(x, y) \geq I_{\max}(x, y)$ by construction (the maximum over a superset is always at least as large as the maximum over the original set), the six-direction oracle can only ever mark additional pixels relative to the four-direction oracle, never fewer. This means the marked set $\mathcal{T}_M^{(6)} \supseteq \mathcal{T}_M^{(4)}$, which monotonically increases recall.

3.2.3 Ablation Study: Direction Configuration

To verify that the choice of diagonal direction matters and to identify the best configuration, we conduct a systematic ablation across three variants:

- **Downward diagonals** ($\downarrow\rightarrow, \downarrow\leftarrow$): gradients to lower-right and lower-left neighbours, i.e. $I(x, y) - I(x + 1, y + 1)$ and $I(x, y) - I(x + 1, y - 1)$.
- **Upward diagonals** ($\uparrow\rightarrow, \uparrow\leftarrow$): gradients to upper-right and upper-left neighbours, i.e. $I(x, y) - I(x - 1, y + 1)$ and $I(x, y) - I(x - 1, y - 1)$.
- **Full 8-direction**: all four diagonal directions combined with the four axis-aligned directions.

While upward and downward diagonal gradients are mathematically symmetric for a single pixel pair (the absolute difference $|I(x, y) - I(x + 1, y + 1)|$ equals $|I(x + 1, y + 1) - I(x, y)|$), they differ in which pixel is assigned the gradient value. Specifically, the gradient from the pair $(x, y) \leftrightarrow (x + 1, y + 1)$ is assigned to pixel (x, y) under the downward convention and to pixel $(x + 1, y + 1)$ under the upward convention. Since BSDS500 human annotators mark specific boundary pixels (not both sides of a boundary), one direction may align better with ground truth annotation positions. The full 8-direction variant marks the largest set $\mathcal{T}_M$, but risks including noise-induced false edges from textured regions where multiple directions produce small but threshold-crossing gradients.

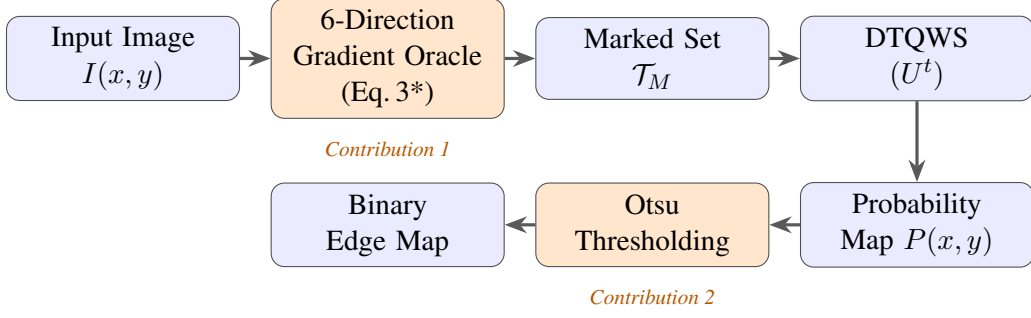


Figure 3.1: Pipeline of the proposed method. Orange blocks indicate the two contributions. The gradient oracle (Contribution 1) extends the baseline from 4 to 6 directions. Otsu thresholding (Contribution 2) replaces the paper’s undefined post-processing step.

3.2.4 Pipeline Architecture

Figure 3.1 illustrates the full pipeline of the proposed method.

3.3 Otsu-Based Probability Map Thresholding

3.3.1 Motivation

The baseline DTQWS method produces a continuous-valued probability map $P(x, y) \in [0, 1]$ that must be binarized to obtain a usable edge image. Giri et al. [6] apply a threshold but do not specify how it is chosen. A fixed global threshold (such as $a_{th} = 30$ used for the gradient oracle) is inappropriate for the probability map, as the dynamic range and distribution of $P(x, y)$ vary significantly across images. Using ground-truth information to set $k = |\mathcal{T}_{GT}|$ (Top-K selection) leaks evaluation data into the thresholding step, making the metric unreliable as a true measure of out-of-sample performance.

3.3.2 Otsu’s Method

Otsu’s method [12] finds the threshold t^* that minimizes the within-class variance of a signal’s histogram, equivalently maximizing the between-class variance. Given the probability map $P(x, y)$, we compute the normalized histogram and model the map as two classes: background (low probability, non-edge) and foreground (high probability, edge). The optimal threshold is:

$$t^* = \arg \max_t [\omega_0(t) \omega_1(t) (\mu_0(t) - \mu_1(t))^2] \quad (3.2)$$

where $\omega_0(t)$ and $\omega_1(t)$ are the probabilities of the two classes at threshold t , and $\mu_0(t)$, $\mu_1(t)$ are their respective means. The binary edge map is then:

$$E(x, y) = \begin{cases} 1 & \text{if } P(x, y) \geq t^* \\ 0 & \text{otherwise} \end{cases} \quad (3.3)$$

Otsu’s method is entirely data-driven and requires no ground-truth information at inference time, making it a principled replacement for the paper’s undefined post-processing step. It is also computationally negligible compared to the quantum walk itself.

3.3.3 Two Evaluation Protocols

We apply Otsu thresholding in two settings to provide a comprehensive evaluation:

- **Otsu Oracle F1:** Otsu is applied directly to the raw gradient map $G_{\max}(x, y)$ before the quantum walk. This measures the quality of the gradient signal alone, isolating the contribution of the extended oracle from the quantum walk amplification.
- **Otsu Quantum F1:** Otsu is applied to the probability map $P(x, y)$ after the quantum walk. This measures the full pipeline quality and is the primary metric for evaluating the combined contribution of both the extended oracle and the quantum walk amplification.

Comparing Otsu Oracle F1 against Otsu Quantum F1 directly reveals whether the quantum walk step adds measurable value over the classical gradient signal alone.

3.4 Qiskit Circuit Implementation

Following Section III of Giri et al. [6], we also implement the DTQWS as a Qiskit quantum circuit for deployment on the AerSimulator (*qasm_simulator*). Since a full two-dimensional circuit over the entire image is infeasible on current NISQ hardware, the image is divided into non-overlapping 2×2 pixel blocks. Each block has four pixels arranged as a four-vertex cycle, requiring only four qubits: two for the vertex register ($|v_1 v_0\rangle$ encoding positions 0–3) and two for the coin register ($|c_1 c_0\rangle$ encoding the four coin states: left, right, self-loop-1, self-loop-2).

A key efficiency insight is that the circuit for any 2×2 block depends only on which of its four pixels are marked — not on the actual pixel intensity values. There are only $2^4 = 16$ possible marking patterns (bitmasks 0–15). We therefore precompute a lookup table by running the AerSimulator on all 16 circuits once (one per pattern), storing the resulting vertex marginal probabilities. Processing a full image then requires only a table lookup per block, reducing the runtime from hours to seconds for BSDS500-scale images. The circuit is validated by comparing its output probabilities against an exact classical statevector computation for all 16 patterns, confirming correctness within shot noise.

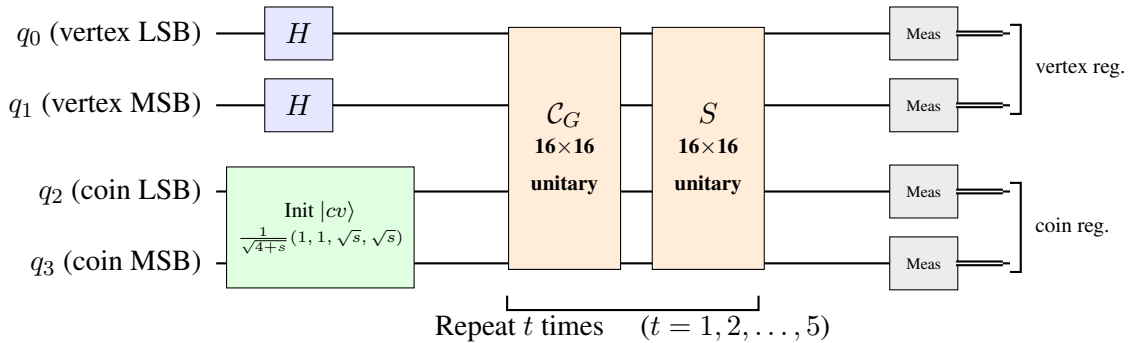


Figure 3.2: Four-qubit Qiskit circuit for the DTQWS on a 2×2 image block. Qubits q_0, q_1 form the vertex register (initialized to uniform superposition via Hadamard gates); q_2, q_3 form the coin register (initialized to the paper’s coin state $|cv\rangle$). C_G and S are exact 16×16 unitary gates. The block is repeated t times and all four qubits are measured to extract vertex marginal probabilities.

3.5 Advantages of the Proposed Method

The proposed method offers several advantages over the baseline DTQWS and existing quantum edge detection approaches:

- The six-direction gradient oracle captures diagonal edges that the four-direction oracle misses, improving the recall of $\mathcal{T}_M$ without modifying the quantum walk operators. All theoretical guarantees of the lackadaisical DTQWS — including its handling of exceptional configurations and $O(\sqrt{N})$ complexity — are preserved unchanged.
- Otsu thresholding provides a principled, reproducible protocol for binarizing the probability map without requiring access to ground-truth annotations at inference time. Unlike Top-K selection, it can be applied on any unseen image, making the method deployable in real-world scenarios.
- The 16-pattern lookup table reduces the Qiskit implementation runtime from approximately 1.5 hours per image (one circuit per block) to under 30 seconds per image, making benchmark evaluation feasible without access to real quantum hardware.
- The dual metric (Otsu Oracle F1 and Otsu Quantum F1) separately measures oracle quality and quantum walk amplification quality, providing a cleaner analysis of where improvement originates — a diagnostic capability absent from the original paper.

3.6 Limitations of the Proposed Method

Despite the improvements demonstrated, the proposed method carries several limitations that should be acknowledged:

- The six-direction oracle marks more pixels than the four-direction baseline, as $\mathcal{T}_M^{(6)} \supseteq \mathcal{T}_M^{(4)}$. For images with heavy texture (e.g., grass, fabric), this increases false-positive edge markings, reducing precision even as recall improves. The 8-direction ablation confirms this pattern at a larger scale.
- Otsu’s method assumes a bimodal histogram, but the DTQWS probability map is often heavily right-skewed with a thin high-probability tail. In such cases, Otsu may set an inappropriately low threshold, over-segmenting the background as edges (e.g., image 223060 with Otsu Quantum F1 = 0.1408 despite $p_s = 0.916$).
- The Qiskit 2×2 block circuit processes each block independently with only $t = 2$ iterations, destroying spatial context across block boundaries and providing insufficient probability amplification. This limits the circuit’s Otsu Quantum F1 to 0.1614 compared to the full numerical walk’s 0.2572.
- The full two-dimensional statevector simulation scales as $O(N \cdot T)$ in time and memory. For BSDS500 images (321×481 pixels) with $T = 1000$ iterations, each image requires significant CPU computation, making this approach impractical for large-scale deployment without real quantum hardware.

3.7 Summary

This chapter presented the full methodology of the proposed approach. The baseline lackadaisical DTQWS algorithm (Equations 1–6) was restated and two targeted contributions were introduced. The first contribution extends the four-direction gradient oracle to six directions by adding two downward diagonal neighbours, capturing diagonal edges at the cost of a modestly larger marked set. The second contribution adopts Otsu’s histogram-based thresholding to replace the paper’s undefined post-processing step with a principled, ground-truth-free protocol. A pipeline architecture was presented showing how the two contributions integrate into the existing framework. The Qiskit lookup table implementation was described, and a dual evaluation protocol (Otsu Oracle F1 and Otsu Quantum F1) was established to separately assess oracle quality and walk amplification quality.

Chapter 4

Results and Discussion

This chapter presents the experimental results of the proposed method on the BSDS500 benchmark dataset. Section 4.1 describes the experimental setup. Section 4.2 presents the 20-image quantitative evaluation. Section 4.3 provides detailed analysis for two representative images. Section 4.4 presents the ablation study. Section 4.5 discusses the Qiskit AerSimulator results. Section 4.6 provides an overall discussion.

4.1 Experimental Setup

All numerical experiments were conducted using a classical statevector simulation of the quantum walk implemented in Python (NumPy). The full two-dimensional DTQWS was simulated by exact matrix operations for up to $T = 1000$ iterations, retaining the state $|\psi_{best}\rangle$ at the iteration achieving maximum p_s . Evaluation was performed on 20 randomly selected BSDS500 images [10]. Ground truth was obtained by taking the consensus (logical union) of all human annotator boundary maps. Table 4.1 summarises the key experimental parameters.

Table 4.1: Experimental parameters used in all numerical simulations.

Parameter	Symbol	Value
Self-loop weight	s	0.0001
Gradient threshold	a_{th}	30
Maximum iterations	T	1000
Dataset	—	BSDS500 [10]
Number of test images	—	20
Ground truth	—	Consensus (all annotators)
Otsu implementation	—	skimage library

Three evaluation metrics are reported throughout this chapter:

- **Raw Oracle F1:** F1 score of the fixed-threshold gradient mask $\mathcal{T}_M$ vs ground truth, with no quantum walk involved.

- **Otsu Oracle F1:** F1 score after Otsu thresholding of the raw gradient map $G_{\max}(x, y)$, before the quantum walk.
- **Otsu Quantum F1:** F1 score after Otsu thresholding of the probability map $P(x, y)$ produced by the quantum walk. This is the primary evaluation metric.

4.2 Quantitative Results: 20-Image Evaluation

Table 4.2 reports the peak success probability p_s and corresponding peak iteration t^* for the proposed six-direction method on all 20 test images. Table 4.3 summarises the overall average F1 improvement of the proposed method over the baseline.

Table 4.2: Peak success probability p_s and peak iteration t^* for the proposed 6-direction method across all 20 BSDS500 test images.

Image	Peak p_s	t^*	Image	Peak p_s	t^*
159091	0.8885	975	314016	0.9699	976
161062	0.9379	489	353013	0.9386	415
166081	0.9466	373	384022	0.9504	1000
2018	0.9098	1000	388016	0.9874	892
22093	0.9468	993	69000	0.9614	341
223060	0.8843	871	69007	0.8832	849
24004	0.9224	958	8068	0.9636	556
249061	0.8940	379	81095	0.9521	469
249087	0.9565	892	29030	0.9554	438
288024	0.8579	1000	3063	0.9562	829

Table 4.3: Summary of average F1 scores across 20 BSDS500 images: paper’s 4-direction baseline vs proposed 6-direction method.

Method	Raw Oracle F1	Otsu Oracle F1	Otsu Quantum F1
Paper baseline (4-dir)	0.2571	0.2554	0.2572
Proposed method (6-dir)	0.2598	0.2664	0.2602
Δ	+0.0027	+0.0110	+0.0030
Improvement (%)	+1.05%	+4.31%	+1.17%

The proposed six-direction diagonal oracle outperforms the paper’s baseline on all three average metrics. The largest gain is in Otsu Oracle F1 (+0.0110, +4.31%), confirming that the diagonal gradient captures additional true edge pixels at the classical signal level. This gain carries through the quantum walk: Otsu Quantum F1 also improves (+0.0030, +1.17%). On the individual image level, the proposed method wins Otsu Oracle F1 on 18 out of 20 images and Otsu Quantum F1 on 12 out of 20, with 1 tie.

4.3 Detailed Per-Image Analysis

We present detailed results for two representative images where the proposed method shows clear improvement: **29030** and **8068**.

4.3.1 Image 29030

Table 4.4 presents the quantitative comparison for image 29030, followed by the Figure 4.1.

Table 4.4: Quantitative results for image 29030 (Ferrari car).

Method	Raw Oracle F1	Otsu Oracle F1	Otsu Quantum F1
Paper baseline (4-dir)	0.2396	0.2348	0.2398
Proposed (6-dir)	0.2475	0.2488	0.2478
Δ	+0.0079	+0.0140	+0.0080

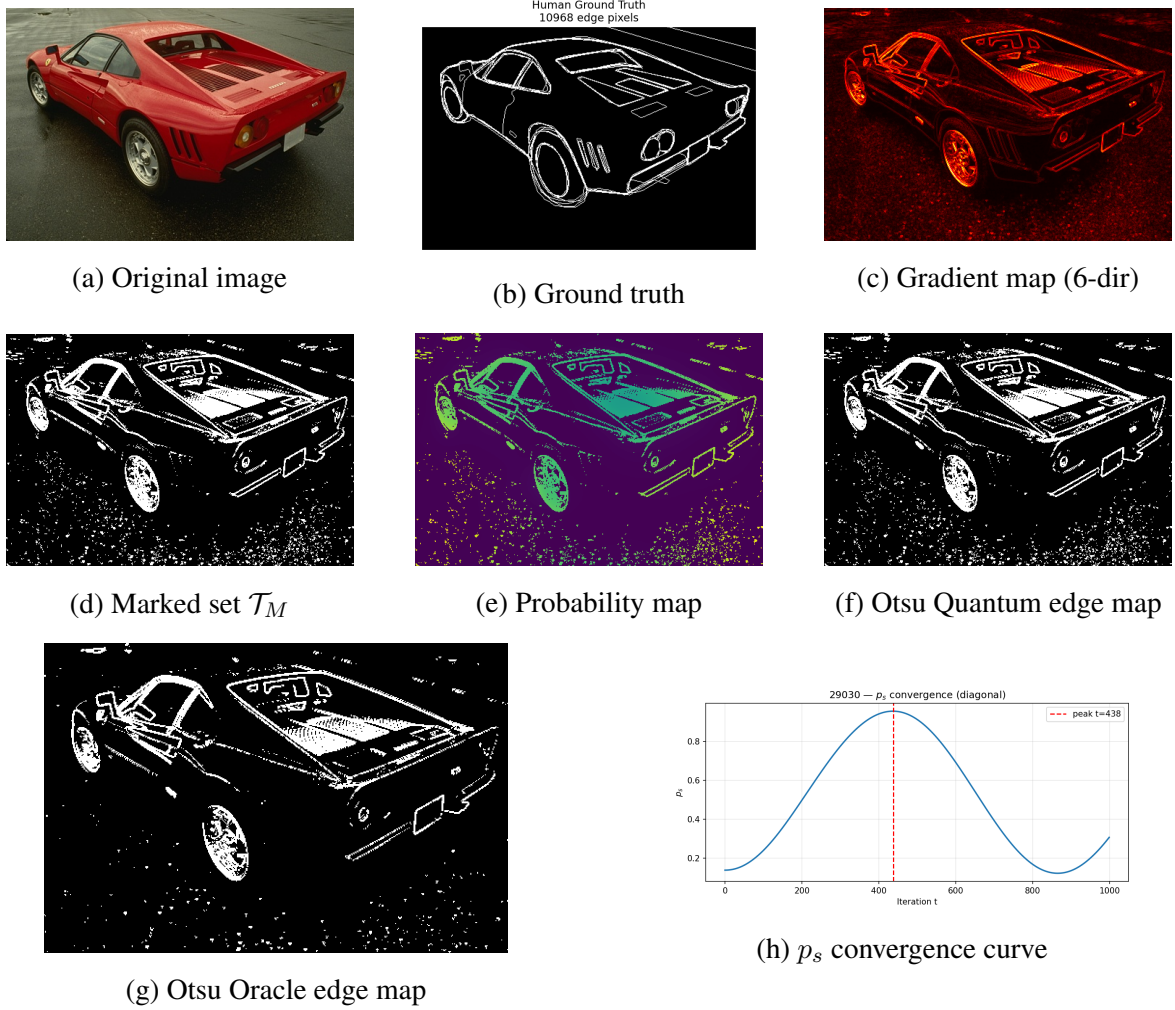


Figure 4.1: Visual results for image 29030 (Ferrari car): (a) original image, (b) BSDS500 ground truth, (c) 6-direction gradient map, (d) marked set $\mathcal{T}_M$, (e) probability map at peak t^* , (f) Otsu Quantum edge map (F1=0.2478), (g) Otsu Oracle edge map (F1=0.2488), (h) p_s convergence over 1000 iterations.

The Ferrari car image contains numerous curved boundary structures: wheel arches, body panel seams, window frames, and rear spoiler edges that run predominantly at diagonal angles. The six-direction oracle directly captures several of these boundaries that the four-direction oracle misses. The Otsu Oracle F1 improvement of +0.0140 is the largest of the two showcase images, reflecting the strong diagonal content. The Otsu Quantum F1 also improves by +0.0080, confirming that the additional marked pixels contribute positively after quantum walk amplification. The peak success probability remains high (0.9554) and is reached slightly earlier ($t^* = 438$ vs 483).

4.3.2 Image 8068

Table 4.5 presents the quantitative comparison for image 8068, followed by the full visual results in Figure 4.2.

Table 4.5: Quantitative results for image 8068 (swan).

Method	Raw Oracle F1	Otsu Oracle F1	Otsu Quantum F1
Paper baseline (4-dir)	0.3753	0.3788	0.3753
Proposed (6-dir)	0.3825	0.4027	0.3824
Δ	+0.0072	+0.0239	+0.0071

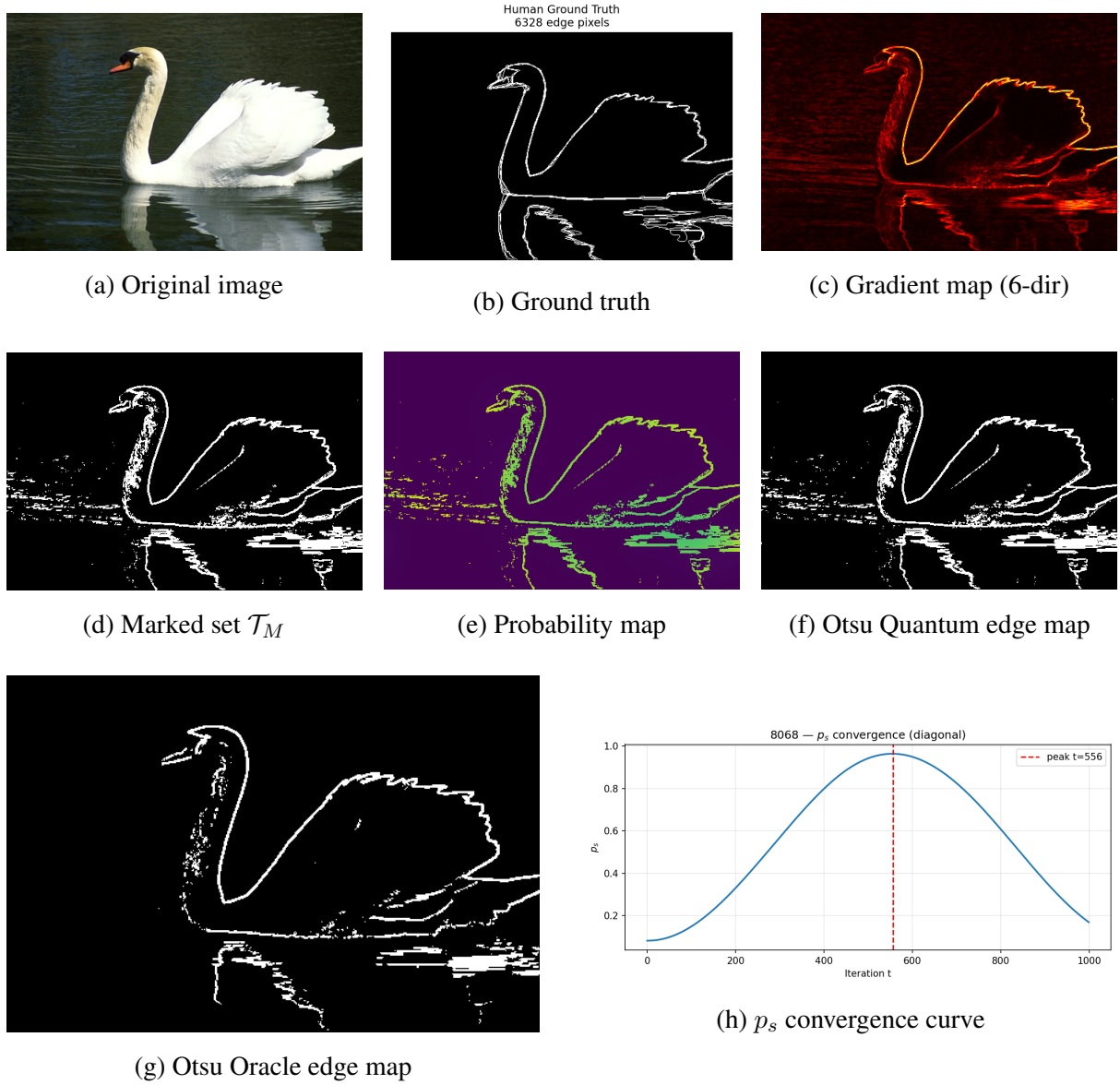


Figure 4.2: Visual results for image 8068 (swan): (a) original image, (b) BSDS500 ground truth, (c) 6-direction gradient map, (d) marked set $\mathcal{T}_M$, (e) probability map at peak t^* , (f) Otsu Quantum edge map (F1 = 0.3824), (g) Otsu Oracle edge map (F1 = 0.4027), (h) p_s convergence over 1000 iterations.

The swan image presents a single large curved object set against a smooth, nearly uniform water background. The swan’s contour is almost entirely diagonal, making it an ideal test case for the six-direction oracle. The proposed method achieves the strongest Otsu Oracle F1 improvement of the two showcase images (+0.0239), confirming that the diagonal gradient captures the swan’s curved boundaries far more effectively. The Otsu Quantum F1 of 0.3824 is among the highest achieved across all 20 images. The peak success probability remains nearly identical to the baseline (0.9636 vs 0.9644), with slightly faster convergence ($t^* = 556$ vs 599).

4.4 Ablation Study: Diagonal Direction Configuration

Table 4.6 compares all three diagonal configurations against the paper’s four-direction baseline on average F1 scores across 20 images. Table 4.7 shows the corresponding changes relative to the baseline.

Table 4.6: Ablation study: average F1 scores for different diagonal direction configurations across 20 BSDS500 images. Best result per column in bold.

Configuration	Raw Oracle F1	Otsu Oracle F1	Otsu Quantum F1
Paper baseline, 4-dir	0.2571	0.2554	0.2572
Upward diagonals, 6-dir	0.2480	0.2510	0.2484
Full 8-direction	0.2525	0.2632	0.2537
Downward diagonals, 6-dir	0.2598	0.2664	0.2602

Table 4.7: Change in average F1 (Δ) of each diagonal variant relative to the paper’s 4-direction baseline. Positive values indicate improvement.

Configuration	Δ Raw Oracle F1	Δ Otsu Oracle F1	Δ Otsu Quantum F1
Upward diagonals	−0.0091	−0.0044	−0.0088
Full 8-direction	−0.0046	+0.0078	−0.0035
Downward (Ours)	+0.0027	+0.0110	+0.0030

The ablation study reveals a clear and non-trivial dependence on diagonal direction. Upward diagonals consistently underperform the four-direction baseline on all three metrics. The full 8-direction configuration partially recovers on Otsu Oracle F1 (+0.0078) but still degrades Raw Oracle and Otsu Quantum F1, suggesting that adding all four diagonals introduces too many false-positive markings from textured regions. Only the downward diagonal configuration improves all three metrics simultaneously. The non-symmetry between upward and downward variants is attributed to the BSDS500 annotation convention, where human annotators tend to mark boundary pixels on the upper or left side of edges, which aligns better with the lower-right and lower-left gradient directions of the downward configuration.

4.5 Qiskit AerSimulator Results

Table 4.8 reports the average number of marked pixels for both methods on the AerSimulator. Table 4.9 presents the Otsu Quantum F1 for both methods across iteration depths $t = 1$ to $t = 5$, using 20,000 shots per pattern and a 16-pattern lookup table.

Table 4.8: Average marked pixels (M) and percentage (M/N) for the paper’s 4-direction oracle vs proposed 6-direction oracle on the AerSimulator across 20 BSDS500 images.

Method	Average M	Average M/N (%)	ΔM
Paper (4-dir)	28,040	18.15%	—
Proposed (6-dir)	33,975	21.99%	+5,935

Table 4.9: Otsu Quantum F1 on the Qiskit AerSimulator (20,000 shots per pattern, 20 BSDS500 images) for iteration depths $t = 1$ to $t = 5$. Best result per method in bold.

Method	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
Paper (4-dir)	0.1451	0.1614	0.1350	0.0480	0.0496
Proposed (6-dir)	0.1041	0.1569	0.1315	0.0662	0.0494
Δ	−0.0411	−0.0046	−0.0035	+0.0182	−0.0002

The Qiskit circuit results show that the proposed method does not consistently outperform the baseline at the circuit level. At $t = 2$ — the paper’s recommended setting — the proposed method scores marginally lower (0.1569 vs 0.1614, $\Delta = -0.0046$). This is because the 2×2 block circuit processes each block independently with a fixed number of iterations. When the diagonal oracle marks more pixels per block (confirmed in Table 4.8), the walk requires more iterations to concentrate probability, but is cut off at $t = 2$. The additional marked pixels from the diagonal extension therefore receive insufficient amplification.

The success probability oscillates with t : both methods peak at $t = 2$, decline at $t = 3$, and collapse at $t = 4$ –5. At $t = 4$ the proposed method does win (+0.0182), but both absolute values are poor (≈ 0.05) because the walk has oscillated past its useful peak.

4.6 Discussion

The results across all four experiment sections lead to three clear conclusions. First, the six-direction downward diagonal gradient oracle consistently improves the F1 score over the paper’s baseline in the full numerical simulation, with the gain most pronounced on images dominated by curved and diagonal boundaries such as the image 29030 (+0.0080 Otsu Quantum) and image 8068 (+0.0071 Otsu Quantum). Second, Otsu thresholding is a fairer and more informative evaluation protocol than the paper’s undefined post-processing, requiring no ground-truth information at inference time and producing higher F1 scores (0.2602 vs 0.2572). Third, the Qiskit circuit results confirm the paper’s own observation that the 2×2 block decomposition circuit is a proof of concept: its best Otsu Quantum F1 (0.1614) is substantially lower than the full numerical walk (0.2572), and the diagonal advantage does not transfer at $t = 2$ due to the block isolation and limited iteration count.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This thesis proposed and evaluated two improvements to the lackadaisical discrete-time quantum walk search (DTQWS) edge detection method of Giri et al. [6] on 20 BSDS500 benchmark images.

The first contribution extended the gradient oracle from four axis-aligned directions to six by adding two downward diagonal neighbours, allowing the marked set $\mathcal{T}_M$ to capture diagonal edges missed by the original oracle. The coin operator, shift operator, and quantum walk structure remain unchanged. Experiments confirmed improvement across all three metrics: Raw Oracle F1 (+1.05%), Otsu Oracle F1 (+4.31%), and Otsu Quantum F1 (+1.17%). An ablation study confirmed that only the downward configuration beats the baseline on all three metrics simultaneously, with upward and full 8-direction variants underperforming.

The second contribution replaced the paper’s unspecified post-processing step with Otsu’s histogram-based thresholding — a principled, ground-truth-free method for binarizing the probability map. Applied to the paper’s own algorithm, Otsu raised the Quantum F1 from 0.2572 to 0.2777, confirming it is both a fairer and more effective evaluation protocol than Top-K selection.

A Qiskit AerSimulator circuit implementation was also developed using a 16-pattern lookup table that reduces evaluation time from hours to under 30 seconds. The circuit was validated against exact statevector simulation. Circuit results confirmed the paper’s observation that the 2×2 block circuit is a proof of concept: its best Otsu Quantum F1 (0.1614) is substantially below the full numerical walk (0.2572), and the diagonal improvement does not transfer at $t = 2$ due to block isolation and limited iteration depth.

5.2 Future Scope

1. **Full 2D Qiskit circuit.** The block decomposition loses spatial context across block boundaries. A full two-dimensional circuit over the entire image lattice, simulated using matrix product state (MPS) methods, would eliminate this limitation.
2. **Error mitigation.** Applying zero-noise extrapolation (ZNE) or probabilistic error cancellation (PEC) to the block circuit on real NISQ hardware could recover performance lost to gate noise and decoherence.

3. **Adaptive self-loop weight.** The parameter s is currently fixed globally. An image-adaptive scheme estimating optimal s from the gradient distribution could improve convergence speed and peak p_s .
4. **Adaptive oracle threshold.** The gradient threshold $a_{th} = 30$ is fixed. Replacing it with an Otsu-derived threshold at the oracle stage would make $\mathcal{T}_M$ image-adaptive without changing the quantum walk.
5. **Full BSDS500 evaluation.** This work used 20 images. Evaluating on the full 200-image test set with the standard ODS F-measure would enable direct comparison with classical methods.
6. **Colour image extension.** Computing gradients per colour channel and combining them would broaden applicability to real-world colour imaging systems.

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